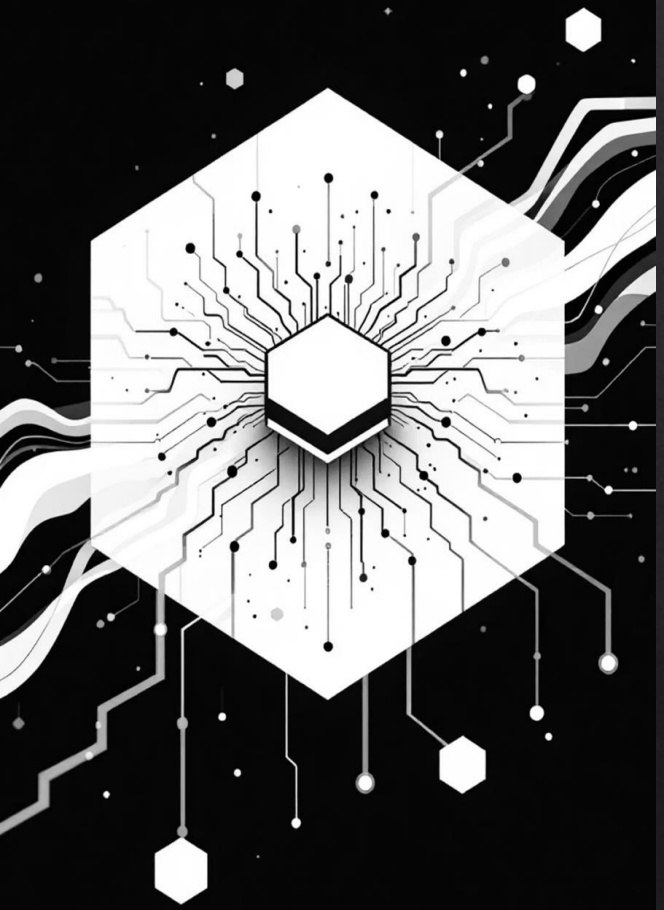


Asking Questions and Tidy Data

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Asking Questions

Why Questions Come First

- **Provides Direction**
  - Defines what to analyze (and what to ignore)
  - Prevents "analysis paralysis" and wasted effort
- **Determines Your Toolkit**
  - Guides what data to collect and how to clean it
  - Chooses the right analytical methods and visualizations
- **Creates Meaningful Outcomes**
  - Frames results into a compelling narrative
  - Ensures insights are actionable and relevant

Types of Analytical Questions

- **Descriptive Questions** - What happened?
- **Diagnostic Questions** - Why did it happen?
- **Predictive Questions** - What will happen?
- **Prescriptive Questions** - What should we do?

Types of Analytical Questions

- **Descriptive Questions (The "What")**
  - **Purpose:** To summarize and describe what has already happened.
  - **Complexity:** Lowest. It's about reporting facts.
  - **Example:** "What were our total sales last quarter?“
- **Diagnostic Questions (The "Why")**
  - **Purpose:** To understand the causes and reasons behind a past event.
  - **Complexity:** Medium. It involves investigation and correlation.
  - **Example:** "Why did sales in the Northeast region drop by 15% in March?"

Types of Analytical Questions

- **Predictive Questions (The "What Will")**
  - **Purpose:** To forecast what is likely to happen in the future based on historical patterns.
  - **Complexity:** High. It often uses statistical and machine learning models.
  - **Example:** "What will our sales be for the next six months?"
- **Prescriptive Questions (The "What Should")**
  - **Purpose:** To recommend a course of action to achieve a desired outcome.
  - **Complexity:** Highest. It combines prediction with optimization and decision theory.
  - **Example:** "What is the optimal mix of marketing channels to maximize sales revenue for the upcoming product launch?"

Question Framework

- Comparison: "How does A compare to B?“
  - Example: "How do sales in Region A compare to Region B?"
- Trend: "How has X changed over time?“
  - Example: "How have monthly sales changed over the past two years?"
- Relationship: "How are Y and Z connected?“
  - Example: "What is the relationship between marketing spend and website traffic?"
- Distribution: "What is the spread/pattern of W?“
  - Example: "What is the distribution of customer ages in our database?
- Composition: " how a whole is divided into its constituent parts. "
  - Example: "What percentage of total sales comes from each product category?"

Common Question Pitfalls

- **Leading Questions:** That suggest a particular answer
- **Vague Questions:** That can't be clearly answered
- **Multiple Questions:** Bundled together
- **Unanswerable Questions:** With current data
- **Irrelevant Questions:** That don't address core goals

Characteristics of Good Questions

Highly effective questions are SMART questions:

- **Specific:** Clear and well-defined
- **Measurable:** Can be answered with data
- **Actionable:** Leads to meaningful insights
- **Relevant:** Addresses real business needs
- **Timely:** Can be answered within constraints

Question Funnel Approach

- **Broad Questions:** General exploration
- **Focused Questions:** Specific areas of interest
- **Precise Questions:** Exact metrics and comparisons

Another Example

- Question: “What products should we promote?"
- SMART Question: "In the last 6 months, which product from the 'Electronics' category has both a sales growth of over 20% and a customer satisfaction score above 4.5 stars?"
  - Specific: Defines the product category and the two key criteria.
  - Measurable: Uses clear metrics (20% growth, 4.5-star rating).
  - Actionable: This product is a prime candidate for a featured promotion.
  - Relevant: Identifies high-performing, well-loved products.
  - Time-bound: Analyzes the last 6 months.

Questions Lead to Data Requirements

- Each question implies data needs
- Questions help identify data gaps
- Questions determine data cleaning requirements
- Questions guide data collection strategy

From Questions to Data Needs

- How questions define your data requirements
- Example: Trend questions need date/time variables
- Example: Comparison questions need categorical variables

The problem in the Data Science world

1. There is a lack of trust from the C-suites (CEO, CFO, CTO etc.) / shareholders / management towards the judgement or results shown by the Data Scientist.
2. As a Data Scientist / Data Analyst, your job is not only to make nice visualizations / charts / graphs / plots, although a lot of the times, you'll be doing this anyway in your effort to explain to the upper management.
3. A lot of CEOs, CFOs complain that 'the data/analysis does not solve a business problem'
4. Then you ask, "what is considered a business problem?"
    a. Make MORE PROFIT (Increase sales, Reduce Cost)
    b. Make MORE PROFIT faster (Reduce waste)

Questions From Data

1. You are given a salary dataset. What kind of business questions can you ask / problems can you solve?
    - Ans: As a CEO, I would be interested in how to pay my employee as competitively as legally possible in the market? (Reduce cost: Human resources, Increase Profit: Good talents can bring good profit)
2. You are given a salary dataset. What kind of insights and analysis can you derive?
    - Ans: What are the factors most correlated with salary? On average, which country earns the most?

Questions Without Looking At Data

3. If you are given a problem to solve (by the C suites), what kind of data that you need?

- CEO: "Hey, I want to attract good talents in Sales department, but I don't want to burn my cash to hire them. Can you help?"
- You: "Ok, I'll need a dataset of salary ranges of our company and other companies as well (the market). So I need the salary dataset."

- Analysis result 1.

  The average salary that a salesperson makes in the company: USD 5000

  The average salary that a salesperson makes in the market: USD 5500

  Conclusion: Your company is paying too low. You won't get the best talents.

- Analysis result 2.

  The average salary that a salesperson makes in the company: USD 5500

  The average salary that a salesperson makes in the market: USD 5500

  Conclusion: Your company matches the market. But this won't attract good/best talents. You want to pay a little bit more "above average" say USD 5750.

Tidy Data

What to check given a dataset?

1. Does the dataset have enough rows and columns?
2. What is the ratio of rows and columns?
3. What is the datatype of each column/variable?

What are tidy data?

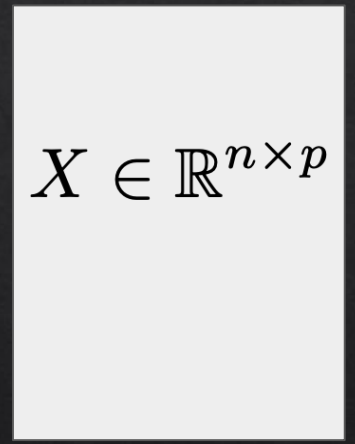
(just renaming, they mean the same thing)

Rules or principles of Tidy data

1. Variables are in columns.
2. Observations are in rows.
3. Each cell contain one value of variable
4. Different observational units types → Different tables

variables

observations



A Simple Example: From Messy to Tidy

Messy Data (Wide Format):

Product	Q1_Sales	Q2_Sales	Q3_Sales
Widget A	210	234	225
Widget B	150	199	188

The variable "Quarter" is spread across multiple columns (Q1_Sales, Q2_Sales, Q3_Sales). This violates rule #1.

A Simple Example: From Messy to Tidy

Tidy Data (Long Format):

Product	Quarter	Sales
Widget A	Q1	210
Widget A	Q2	234
Widget A	Q3	225
Widget B	Q1	150
Widget B	Q2	199
Widget B	Q3	188

Cases:

country	2020	2021
Belgium	14.444	14.791
Bulgaria	23.128	22.305
Czechia	8.610	8.990

Population:

country	2020	2021
Belgium	11.522.440	11.554.767
Bulgaria	6.916.548	6.951.482
Czechia	10.494.836	10.693.939

country	year	rate
Belgium	2020	14.444/11.522.440
Belgium	2021	14.791/11.554.767
Bulgaria	2020	23.128/6.916.548
Bulgaria	2021	22.305/6.951.482
Czechia	2020	8.610/10.494.836
Czechia	2021	8.990/10.693.939

The rate column contains 2 variables (the number of cases and the population count), and as such violates the rules for tidy data.

country_year	cases	population
Belgium_2020	14.444	11.522.440
Belgium_2021	14.791	11.554.767
Bulgaria_2020	23.128	6.916.548
Bulgaria_2021	22.305	6.951.482
Czechia_2020	8.610	10.494.836
Czechia_2021	8.990	10.693.939

Here the country_year column is violating the “1 column = 1 variable” rule (or: cells in the column are violating the “1 cell = 1 value” rule).

Subtlety: Observation Amibuity

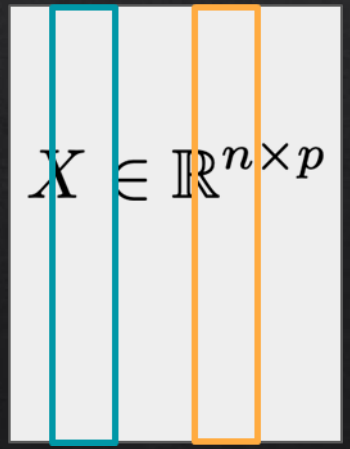
- What are considered variables vs. observations may vary throughout your analysis.
- Rules of thumb
  - Functional relationships are easiest to see through columns
  - Group comparisons are easiest to see through rows
  - An observation is the smallest unit you'd like to draw conclusions or make predictions about

$$X \in \mathbb{R}^{n \times p}$$

Example : "Functional relationships are easiest to see through columns"

- To model Weight as a function of Height and Age, you need a dataset where Weight, Height, and Age are each in their own column.

Height	Age	Weight
175	30	70
160	25	60

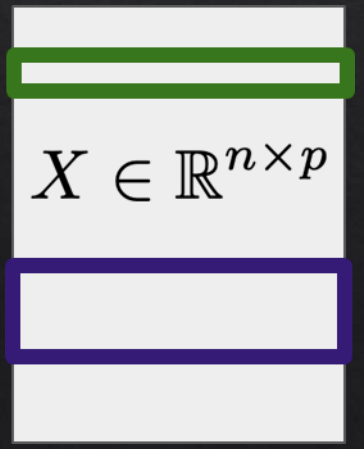


The dependent and independent variables should be in separate columns.

Example : "Group comparisons are easiest to see through rows”

If your goal is to compare different groups (e.g., "Compare the average Blood Pressure of Group_A vs. Group_B"), the group identifier should be in one column, and the measurement you're comparing should be in another. This often means having multiple rows for the same "entity."

Group	Blood_Pressure
Group_A	120
Group_A	118
Group_B	110
Group_B	108



Example: "An observation is the smallest unit you'd like to draw conclusions or make predictions about ”

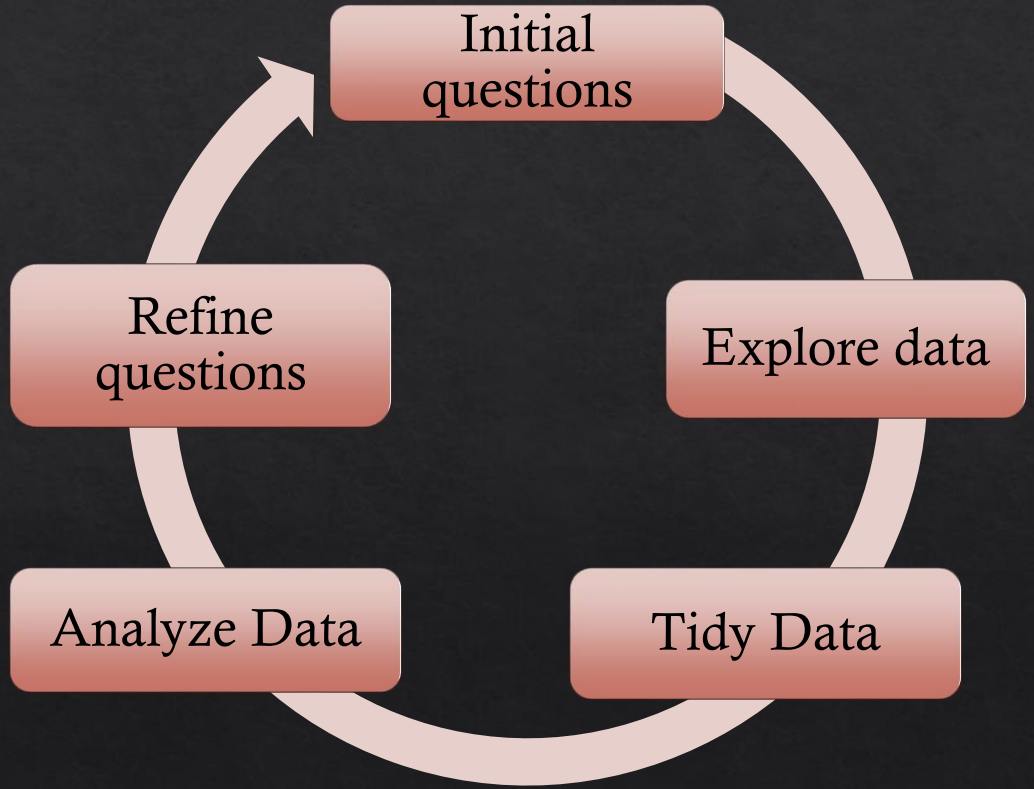
- **Example 1 (Clinical Trial):**
  - **Question:** "Which drug is more effective for patients?"
    - **Observational Unit:** A `Patient`.
    - **Data Structure:** One row per patient, with columns for `patient_id`, `drug_type`, `recovery_time`.
  - **Question:** "How does the effectiveness of this drug change over time for a given patient?"
    - **Observational Unit:** A `Patient` at a specific `Time`.
    - **Data Structure:** One row per patient per time point, with columns for `patient_id`, `drug_type`, `time_point`, `symptom_score`. Now, a single patient appears over multiple rows.

Example : "An observation is the smallest unit you'd like to draw conclusions or make predictions about"

Example 2 (E-commerce):

- **Question:** "What is the total revenue per customer?"
  - **Observational Unit:** A `Customer`.
  - **Data Structure:** One row per customer.
- **Question:** "What products are often purchased together?"
  - **Observational Unit:** A `Transaction` (a single purchase event).
  - **Data Structure:** One row per transaction. A single customer will appear in multiple rows if they have made multiple purchases.

Iterative Question Refinement



Reading Task

- https://faculty.washington.edu/otoomet/datascience-intro/questions-answers.html
- https://towardsdatascience.com/how-to-ask-the-right-questions-as-a-data-scientist-913621907411
- https://commoncog.com/becoming-data-driven-in-business/

Thank You